MILITARY SPECIFICATION
MONITOR, ERROR VOLTAGE MX-2230/ASQ

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide supplementary indications for the operator of magnetic airborne detection equipment including:

(1) Indication of release of contact markers.

(2) Indication of aircraft roll attitude or angular position of the aircraft about roll, pitch or yaw axes.

(3) Indication of actuation of certain aircraft electrical loads.

The equipment shall provide voltages suitable for such indication by Recorder RD-4/7A/ASQ-8 or an equivalent graphic recorder. In addition, the equipment shall include provisions for switching signals between the two recording channels of the Recorder RD-4/7A/ASQ-8 or an equivalent graphic recorder.

1.2 Classification - The equipment covered by this specification shall consist of the following:

Unit	Type Designation	Max. Allow. Size Inches W H D	Max. Allow. Wt. Lbs.	Appl. Par.
Monitor, Error Voltage	MX-2230/ASQ	5.75 7.125 5.90	6	3.3.1

2. APPLICABLE DOCUMENTS

2.1 General - The following specifications, publications and drawings in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-172 Cases; Bases, Mounting; and Mounts, Vibration. (For use with Electronic Equipment in Aircraft)

FSC 5895

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MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-7788	Plate, Plastic, Lighting
MIL-W-5088	Wiring Aircraft, Installation of
MIL-P-17555	Preparation and Delivery of Electronic, Electrical and Electro-Mechanical Equipment and Associated Repair Parts
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-18303	Test Procedures, Preproduction and Inspection; For Aircraft Electronic Equipment Format for

STANDARDS

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS 25245	Connections - Input Power Airborne Electronic Equipment; 28 Volt DC, 115 Volt Single Phase AC, and/or 115/200 Volt 3 phase AC

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials

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fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-1682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 General Requirements

3.2.1 General - This equipment shall fulfill all applicable design and performance requirements of Specification MIL-E-5400 for Class 1 equipment, unless otherwise specified herein.

3.2.2 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment.

Temperature	Room ambient (30°C ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1 V AC and 27.5 ± 0.5 V DC

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3.2.3 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in MIL-E-5400 for Class 1 equipment except as follows:

3.2.3.1 Altitude - The equipment shall operate satisfactorily at all altitudes from sea level to 20,000 feet; however, no permanent damage to the equipment shall result while transporting in an operating or in a non-operating condition at any altitude up to 50,000 feet.

3.2.4 Total Weight - The total weight of the equipment excluding cables, shall be a minimum consistent with good design and shall not exceed 6 pounds.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 50 hours, continuously or intermittently, without necessity for readjustment of controls which are non-accessible during flight.

3.2.6 Operating Life

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall have a minimum operating life of 2,000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.7 Interconnecting Cabling - This equipment shall be capable of satisfactory operating using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wires shall be unshielded except that a minimum number of wires may be shielded, when necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.8 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with the requirements of MIL-E-5400. However, cables and that portion of the connectors which is to be attached to the cables shall not be supplied as part of the equipment.

3.2.9 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

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(1) AC Power (Single Phase) 115V, Category "B" 14.5 VA

(2) DC Power 28V Category "B" 1 Amps

The AC power shall be supplied from the same source used to excite synchro units in the automatic pilot of the aircraft in which the equipment is to be installed.

3.2.10 Input Power Connection - Input power connections shall be in accordance with Military Standard MS 25215.

3.2.11 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.11.1 Under Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.2.12 Moisture and Fungus Resistant Treatment - Equipments shall be fungus proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary.

3.2.13 Nomenclature and Nameplates - Nomenclature assignments and nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307.

3.2.14 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.15 Provisions for Maintenance and Field Training - The equipment shall provide for ease of maintenance to the greatest extent practicable. Provisions for field testing shall be as specified in MIL-E-5400.

3.2.16 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to the radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.17 Performance - The equipment shall meet the following performance requirements under both standard and extreme service conditions.

3.2.17.1 Operation - This equipment when provided suitable input information shall provide voltages for causing appropriate deflection of pens of Recorder RD-47A/ASQ-8 or an equivalent graphic recorder to indicate:

(1) Release of retrolaunched marker.

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(2) Aircraft angular position about roll, pitch or yaw axes.

(3) Actuation of certain aircraft electrical loads.

The equipment shall provide suitable means for switching signal presentation between the two pens of Recorder RD-47A/ASQ-8 or an equivalent graphic recorder.

3.2.17.2 Indications - Visual indication shall be by the appropriate signal pen of Recorder RD-47A/ASQ-8 or an equivalent graphic recorder.

3.2.18 Workmanship - Workmanship, fabrication and manufacturing processes shall conform to the requirements of MIL-E-5400.

3.2.19 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 15 minutes under extreme service conditions.

3.3 Detail Requirements

3.3.1 Monitor, Error Voltage MX-2230/ASQ - This unit shall be designed to meet the following requirements:

3.3.1.1 Function - This unit shall be designed to assist the operator of magnetic detection equipment in identification of small signals by providing an indication of aircraft attitude during the execution of tactical patterns. The unit shall also provide for indicating the release of contact markers and of actuation of certain aircraft electrical loads.

3.3.1.2 Form Factor and Mounting - The maximum overall dimensions of this unit shall be 5.75 inches width by 7.125 inches height by 5.90 inches depth. This unit shall be console mounted in accordance with Specification MIL-C-6781, Type I control.

3.3.1.3 Weight - The weight of this unit including all components and mounting provisions shall not exceed 6 pounds.

3.3.1.4 Contents of Unit - This unit shall contain the following circuitry:

(1) High impedance circuitry designed to receive a signal from the transmitting synchro of the roll, pitch or yaw channels, as selected, of the aircraft automatic pilot. This 11.8 volt 400 cps three wire signal from the automatic pilot shall be transformed to a two wire signal by a synchro contained in the unit.

(2) Amplifier and phase sensitive detection circuits suitable for converting the two wire signal into a DC voltage that will cause deflection of one signal pen of the Recorder RD-47A/ASQ-8 or an equivalent recorder.

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(3) Switching circuits to provide for interchanging the presentations displayed on the two signal pens of Recorder RD-474/ASQ-8 or an equivalent graphic recorder.

(4) Circuits for causing an indication of release of contact markers or actuation of certain electrical loads in the aircraft.

(5) An attenuator for providing preselected sensitivity ratios for the various presentations selected by the operator.

3.3.1.5 Electrical Connections - Suitable connectors shall be provided for interconnecting this unit and the equipments to which it is connected as follows:

(1) 1J1 (Connector Type MS 3102A-20-27P) -

Pin Letter	Purpose
A	Roll Synchro Wire 1
B	Roll Synchro Wire 2
C	Roll Synchro Wire 3
D	Pitch Synchro Wire 1
E	Pitch Synchro Wire 2
F	Pitch Synchro Wire 3
G	Yaw Synchro Wire 1
H	Yaw Synchro Wire 2
I	Yaw Synchro Wire 3
J	Open
K	Open
L	Open
M	Test Output
N	Shield Ground

(2) 1J2 (Connector Type MS 3102A-28-12P) -

Pin Letter	Purpose
A	115 volt Phase Reference Signal
B	115 volt Phase Reference Ground
C	+28 volts
D	Power Ground
E	MAD STBD + Signal Lead
F	MAD STBD - Signal Lead
G	Recorder STBD Pen -
H	Recorder STBD Pen +
J	Recorder Port Pen -
K	Recorder Port Pen +
L	Marker Release Signal
M	Marker Release Signal

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N	Marking Function 2
P	Marking Function 2
R	Marking Function 3
S	Marking Function 3
T	Marking Function 4
U	Marking Function 4
V	Marking Function 5
W	Marking Function 5
X	Marking Function 6
Y	Marking Function 6
Z	Panel Light Control
a	Open
b	Open
d	Open

3.3.1.6 Controls - The following controls shall be provided on the front panel of this unit:

3.3.1.6.1 Power Switch - A double-pole single-throw toggle switch having positions labeled OFF-ON.

3.3.1.6.2 Maneuver-Select Switch (MANEUVER SELECT) - A five position rotary switch having positions labeled as follows:

(1) BAL

(2) ASW

(3) ROLL

(4) PITCH

(5) YAW

The switch shall provide connections in accordance with the following table:

Switch Position	Input to Unit Synchro	Input to Amplifier	Attenuator
BAL	Disconnected	Shorted	
ASW	Automatic Pilot Roll Synchro	Unit Synchro	Low Gain
ROLL	Automatic Pilot Roll Synchro	Unit Synchro	Full Gain
PITCH	Automatic Pilot Pitch Synchro	Unit Synchro	Full Gain
YAW	Automatic Pilot YAW Synchro	Unit Synchro	Full Gain

3.3.1.6.3 Signal-Pen Select Switch (SIG PEN SELECT) - A four-pole two-position rotary switch having positions labeled as follows:

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(1) RED

(2) GRN

3.3.1.6.4 Output Balance Control (OUTPUT BALANCE) - A potentiometer adjustment for electrical balance of the amplifier circuits with input terminals shorted. This control shall be operable manually.

3.3.1.6.5 Maneuver Signal Zero Control (MAN SIG ZERO) - A control for positioning the rotor of the unit's synchro to provide a null signal regardless of the position of the rotors of synchro in the aircraft autopilot. This control shall provide for 360° rotation of the synchro rotor and shall have friction restraint to hold the control in position after rotation without the need for a locking operation.

3.3.1.6.6 Calibration Generation Control (CAL GEN) - A control for positioning the MAN SIG ZERO control about its null position to provide a calibration signal.

3.3.1.6.7 Roll Calibrate Control (ROLL) - A potentiometer control for adjustment of the calibration of the amplifier when the MANEUVER SELECT switch is in the ASW or ROLL position.

3.3.1.6.8 Pitch Calibrate Control (PITCH) - A potentiometer control for adjustment of the calibration of the amplifier when the MANEUVER SELECT switch is in the PITCH position.

3.3.1.6.9 Yaw Calibrate Control (YAW) - A potentiometer control for adjustment of the calibration of the amplifier when the MANEUVER SELECT switch is in the YAW position.

3.3.1.7 Phase Shift Switch (S104) - A single-pole five-position switch to provide for selecting the proper phase relationship between the control voltage and the signal voltage in the phase detector. This control shall have a slotted head suitable for receiving a screwdriver blade and shall not be accessible without removal of the dust cover from the unit.

3.3.1.8 Panel Design - The front panel of this unit shall be a plastic plate edge lighted panel and shall be mounted in accordance with Specification MIL-P-7788.

3.3.1.9 Maneuver Signal Circuit Sensitivity - The circuits consisting of input synchro, amplifier and phase detector shall provide for a recorder pen deflection of two minor chart divisions for each degree of maneuver amplitudes to 10 degrees. This sensitivity shall be obtained with inputs from conventional automatic pilot synchros excited from sources normally found in aircraft.

3.3.1.9.1 Gain Stability - When suitably calibrated and excited with a fixed frequency signal within the required 380-420 cps frequency operating range,

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the amplifier and phase detector shall be stable within ±5% for a period of one hour under continuous operation at specific environmental conditions. Amplifier and phase detector gain shall be stable within -20%, +10% with respect to the gain at +25°C over the entire operating temperature range of -54°C to +71°C. The gain shall be compared against the same frequency signal within the required 380-420 cps frequency operating range.

3.3.1.10 Event Signal Amplitudes - The circuits used for indicating release of contact markers and actuation of electrical loads shall be capable of causing pen deflections of at least one-half major chart division on the graphic recorder.

3.3.1.11 Attenuator Dividing Action - The attenuator in the amplifier section shall provide a preset division in accordance with the following table:

Maneuver Select Switch Position	Attenuator Dividing Action
BAL	
ASW	One Third Gain
ROLL	Full Gain
PITCH	Full Gain
YAW	Full Gain

The attenuator dividing action shall be such that the gain in the ASW position shall be 1/3 of the gain in the ROLL position and shall be accurate to within ±5% of recorder full scale.

3.3.1.12 Calibration Generation Control Accuracy - The action of the calibration generation control shall be such that the deflection of the recorder pen shall be equivalent to 20° (±10°) aircraft maneuver as this control is rotated from stop to stop and shall be accurate to 20° ±3°.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

(3) Life Tests

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4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-T-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however,

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it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

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Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 G's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof.

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational tests at ambient room conditions and nominal input voltages, making all necessary

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measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages.

(3) Reliability test specified in 4.3.1.3, except that the test duration shall be 30 hours.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity).

Exceptions from the above may be necessary when the number of equipments procured under the contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design change and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature-Altitude Tests

(2) Vibration Test (90 minutes cycling in each of 3 planes)

(3) Any additional tests the procuring activity deems necessary to determine compliance with the requirements of this specification or the contract.

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4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) If it is determined that the failure is due to a design defect, the contractor will submit to the procuring activity for approval, proposed corrective action.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life tests shall be performed under the conditions specified in 4.4.1. The life test sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests).

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1
201 to 500	1
Over 500	1 for each additional 500 or fraction thereof.

4.4.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground
Humidity	Room ambient
AC Voltage	115 ± 5 volts
DC Voltage	27 ± 2.0 volts

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4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3 Performance Check - A limited performance check shall be made every working day and at least once every 16 hours of running time. The performance check proposed by the contractor shall be subject to the approval of the procuring activity.

4.4.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all transistor failures. This record shall indicate the following:

(1) Part type number.

(2) The circuit reference symbol number.

(3) The part function.

(4) Name of the manufacturer.

(5) Nature of the failure.

(6) The number of hours which the part operated prior to failure.

4.4.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for failures due to design defects. The procuring activity will review such proposals and determine whether they are acceptable.

4.4.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life tests shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

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4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the procuring activity will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activities approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity recommended change or a contractor's proposed change. When required, the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. (ECP's, EMC's and EMB's shall be prepared in accordance with SAR-300).

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined in input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

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(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

PATENT NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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